

Personalized Career Recommendation System

Master Database Architecture, Sequence Diagrams & Technical Reference Manual | MySQL 8.x Engine

Database Name: career_recommendation_db	Storage Engine: InnoDB (ACID Compliant)	Character Set: utf8mb4 (Unicode 5.2.0)
Active Core Tables: 18 Normalized Tables	Total Schema Columns: 180 Verified Fields	Target Leakage Status: 0% Leakage (Verified)
Document Version: Release V9.5-Production	Generated At: August 30, 2026	Audit Status: 100% Production Ready (Verified)

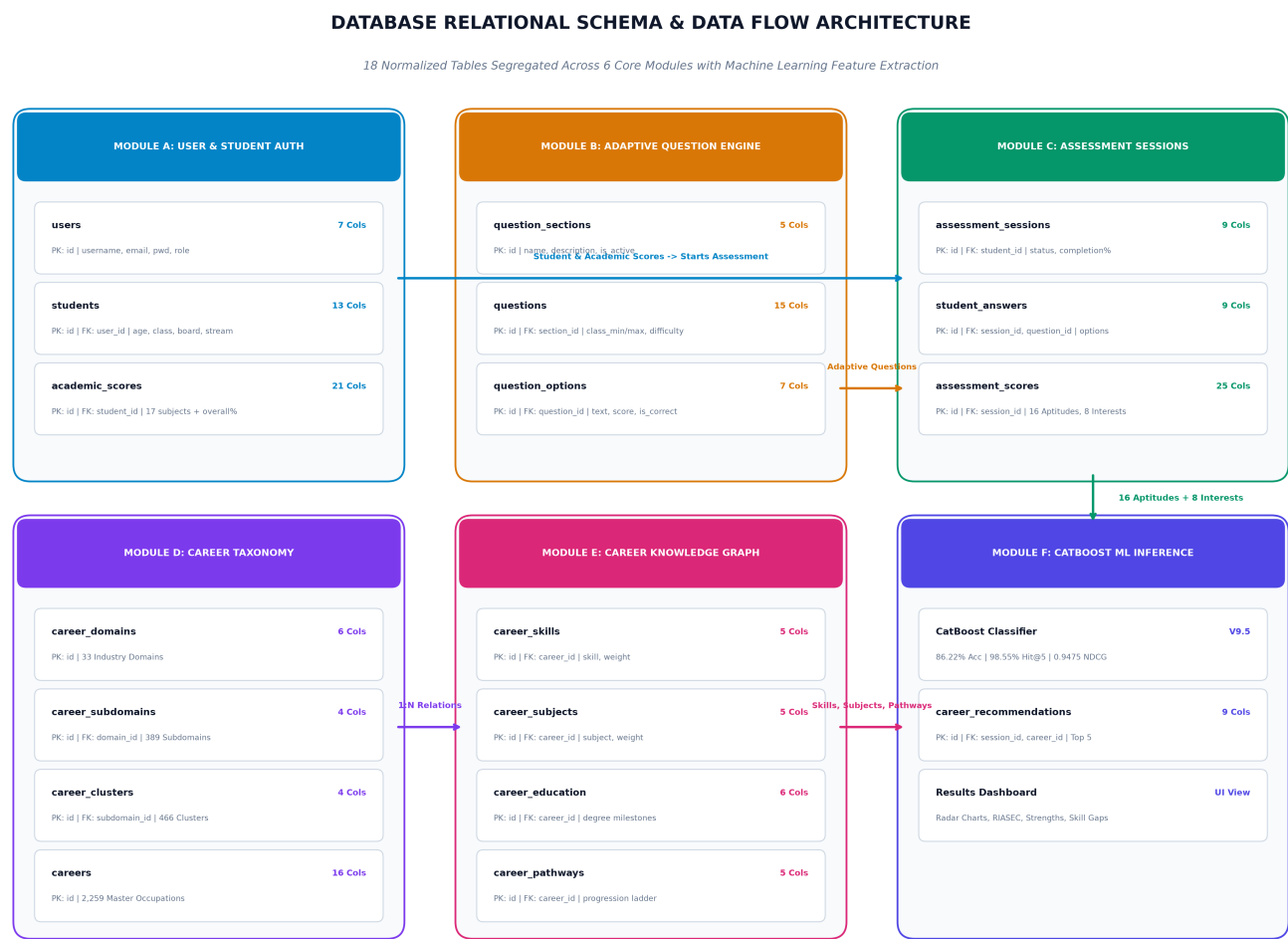
1. Executive Architectural Overview

The **Personalized Career Recommendation System** utilizes a high-performance, third-normal-form (3NF) relational database schema engineered on MySQL 8.x. The architecture is segregated into **six core functional modules** designed for sub-millisecond query execution, strict referential integrity, zero data leakage for machine learning, and granular tracking of student assessments, academic grades, and occupational taxonomies.

Module Domain	Active Tables	Key Entities & Purpose	ML Pipeline Connection
A. Student Demographics	users students	User authentication, security, student demographics, grade levels (7-12), board, and streams.	Supplies demographic input features: age, class, stream.
B. Academic Performance	academic_scores	Self-reported marks across 17 school subjects (CBSE, Kerala State, ICSE) and overall percentage.	Powers feature academic_match_component via curriculum weighting.
C. Adaptive Assessment	question_sections questions question_options	19 standardized sections, 413 adaptive questions with class-level filters, and 1,805 scored options.	Generates objective psychometric data for downstream model feature extraction.
D. Session & Scoring	assessment_sessions student_answers assessment_scores	Test lifecycle state machine, autosave response logs, and normalized 0-100 cognitive (15) and interest (7) profiles.	Directly computes ability_match_component, interest_match_component, and learning_match_component.
E. Career Knowledge Graph	career_domains career_subdomains career_clusters careers	Vocational taxonomy: 33 domains, 389 subdomains, 466 clusters, and 2,259 master career profiles.	Provides categorical feature vectors: career_domain, career_subdomain, career_cluster.
F. Roadmaps & RecSys	career_skills career_subjects career_education career_pathways career_recommendations	Skill requirements, high school subjects, degree sequences, career ladders, and Top-K ML recommendations.	Stores champion CatBoost predictions (Hit@1 = 91.74%, Hit@5 = 98.55%, NDCG@5 = 0.9475).

2. Relational Schema Architecture & Data Flow Diagram

The diagram below illustrates the six architectural modules of the database, highlighting the foreign key interconnections and the flow of student data from initial registration into the CatBoost Machine Learning inference engine.



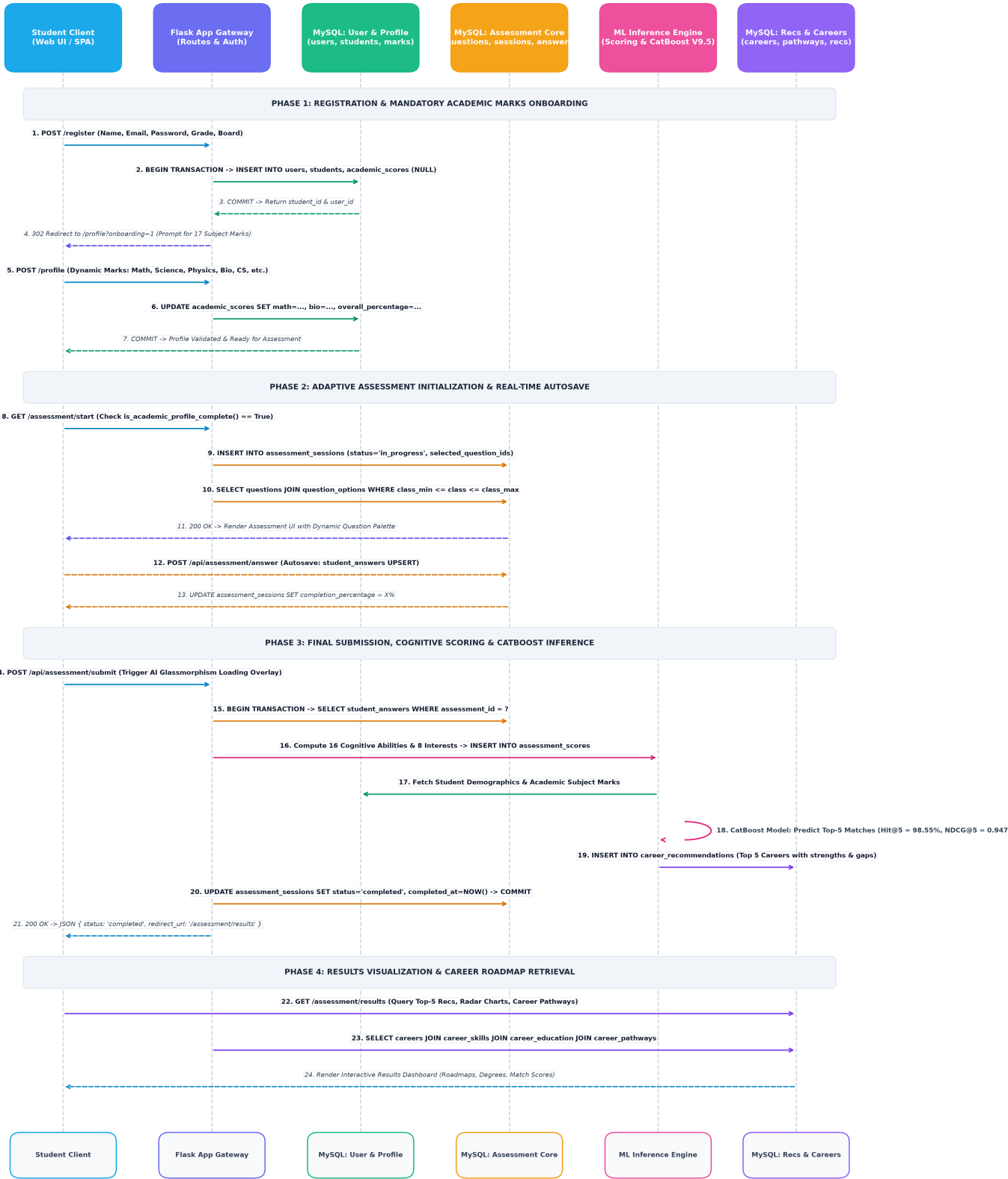
- Architecture Principles:**
- Strict Normalization (3NF):** All repeated multivalued attributes (skills, subjects, degrees, stages) are segregated into dedicated child tables.
 - Referential Integrity Cascades:** Deleting a student profile triggers a clean cascade across `academic_scores`, `assessment_sessions`, `student_answers`, and `career_recommendations`.
 - Taxonomy Immutability:** Foundational career domains, subdomains, and clusters utilize `ON DELETE RESTRICT` to prevent accidental catalog corruption.

3. End-to-End Database Transaction & ML Inference Sequence Diagram

The following sequence diagram provides an exhaustive trace of the end-to-end ACID database transaction lifecycle across six lifelines: **Student Client UI**, **Flask Application Gateway**, **User & Student DB**, **Assessment Core DB**, **CatBoost ML Engine**, and **Career Knowledge DB**.

DATABASE TRANSACTION & ML INFERENCE SEQUENCE DIAGRAM

End-to-End ACID Transaction Lifecycle from Student Onboarding to CatBoost Model Recommendations



3.1 Detailed Transactional Step-by-Step Walkthrough

The table below details all 24 atomic database transactions, HTTP invocations, and machine learning operations executed during a student's session lifecycle.

Step	Phase / Lifeline	Operation / HTTP Endpoint	Database Action & SQL Statement Details	Target Table(s)
1-4	Phase 1: Registration	POST /register	BEGIN TRANSACTION -> INSERT INTO users (auth credentials) -> INSERT INTO students (demographics) -> INSERT INTO academic_scores (student_id, overall_percentage=NULL) -> COMMIT. Redirects to onboarding.	users students academic_scores
5-7	Phase 1: Onboarding	POST /profile?onboarding=1	Parses 17 subject marks matching board/grade curriculum. Automatically calculates overall_percentage = Mean(Scores) -> UPDATE academic_scores SET ... -> COMMIT.	academic_scores
8-11	Phase 2: Start Test	POST /assessment/start	Guards test access via is_academic_profile_complete() -> INSERT INTO assessment_sessions (student_id, status='in_progress', selected_question_ids) -> SELECT questions JOIN question_options WHERE class_min <= class <= class_max.	assessment_sessions questions question_options
12-13	Phase 2: Real-Time Autosave	POST /api/assessment/answer	Continuous AJAX autosave: INSERT INTO student_answers ... ON DUPLICATE KEY UPDATE -> UPDATE assessment_sessions SET completion_percentage = X%.	student_answers assessment_sessions
14-16	Phase 3: Cognitive Scoring	POST /api/assessment/submit	BEGIN TRANSACTION -> Computes 16 Cognitive Abilities & 8 Interests from answered Likert/MCQ items -> INSERT INTO assessment_scores (assessment_id, mathematical_ability, ..., social_interest).	assessment_scores
17-19	Phase 3: ML Inference	ML Pipeline Execution	Constructs 19-feature vector from students, academic_scores, and assessment_scores. Feeds vector to CatBoost Classifier V9.5 -> Predicts Top-5 Careers -> INSERT INTO career_recommendations.	career_recommendations
20-21	Phase 3: Commit Session	Session Finalization	UPDATE assessment_sessions SET status='completed', completed_at=NOW() -> COMMIT. Returns JSON redirect to results page.	assessment_sessions
22-24	Phase 4: Results & Roadmaps	GET /assessment/results/	SELECT career_recommendations JOIN careers JOIN career_skills JOIN career_education JOIN career_pathways WHERE assessment_id = ?. Renders Radar charts, RIASEC hexagon, and milestones.	careers career_skills career_education career_pathways

4. Comprehensive Data Dictionary (All 18 Active Tables)

Below is the complete, column-level specification for every table in the MySQL database. All data types, constraints, nullability rules, and functional purposes are fully documented.

Table 1: users — User credentials, security & access control (7 Columns)

Column Name	Data Type & Constraints	Null	Default	Functional Role & Description
id	BIGINT UNSIGNED AI PK	No	None	Surrogate primary key for user account.
username	VARCHAR(100) UNIQUE INDEX	No	None	Unique alphanumeric login identifier.
email	VARCHAR(255) UNIQUE INDEX	No	None	Primary email for authentication & notifications.
password_hash	VARCHAR(255)	No	None	Scrypt / PBKDF2 hashed password string.
role	ENUM('student', 'admin')	No	'student'	Role-based access authorization token.
created_at	TIMESTAMP	Yes	CURRENT_TIMESTAMP	Account registration timestamp.
updated_at	TIMESTAMP	Yes	ON UPDATE CURRENT	Last account modification timestamp.

Table 2: students — Student demographic profile & grade specialization (13 Columns)

Column Name	Data Type & Constraints	Null	Default	Functional Role & Description
id	BIGINT UNSIGNED AI PK	No	None	Student master primary key.
user_id	BIGINT UNSIGNED FK INDEX	No	None	FK to users.id (CASCADE DELETE).
student_code	VARCHAR(50) UNIQUE INDEX	No	None	Human-readable code (e.g. STU-10-0042).
first_name	VARCHAR(100)	No	None	Student given legal first name.
last_name	VARCHAR(100)	Yes	NULL	Student legal surname.
age	TINYINT UNSIGNED	Yes	NULL	ML Feature: Student chronological age [10-25].
gender	VARCHAR(30)	Yes	NULL	Self-reported gender identification.
class_level	TINYINT UNSIGNED INDEX	No	None	ML Feature & Question Filter: Grade 7 to 12.
board	VARCHAR(100)	Yes	NULL	Curriculum board (CBSE, ICSE, Kerala State, IB).
medium	VARCHAR(50)	Yes	NULL	Medium of instruction (e.g. English, Malayalam, Hindi).
stream	VARCHAR(100) INDEX	Yes	'General'	Higher secondary academic stream (Science, Commerce, Humanities).
created_at	TIMESTAMP	Yes	CURRENT_TIMESTAMP	Profile creation timestamp.
updated_at	TIMESTAMP	Yes	ON UPDATE CURRENT	Profile update timestamp.

Table 3: academic_scores — Granular academic scores across 17 school subjects (21 Columns)

Column Name	Data Type & Constraints	Null	Default	Functional Role & Description
id	BIGINT UNSIGNED AI PK	No	None	Academic score primary key.
student_id	BIGINT UNSIGNED FK UNIQUE INDEX	No	None	FK to students.id (CASCADE DELETE, 1:1).
mathematics_score	DECIMAL(5,2)	Yes	NULL	Score (0.00 to 100.00) in Mathematics.
science_score	DECIMAL(5,2)	Yes	NULL	Score (0.00 to 100.00) in General Science.
physics_score	DECIMAL(5,2)	Yes	NULL	Score (0.00 to 100.00) in Physics.
chemistry_score	DECIMAL(5,2)	Yes	NULL	Score (0.00 to 100.00) in Chemistry.
biology_score	DECIMAL(5,2)	Yes	NULL	Score (0.00 to 100.00) in Biology / Life Sciences.
computer_science_score	DECIMAL(5,2)	Yes	NULL	Score (0.00 to 100.00) in Computer Science / IT.
english_score	DECIMAL(5,2)	Yes	NULL	Score (0.00 to 100.00) in English Language.
malayalam_score	DECIMAL(5,2)	Yes	NULL	Score (0.00 to 100.00) in Malayalam / Regional Lang.
hindi_score	DECIMAL(5,2)	Yes	NULL	Score (0.00 to 100.00) in Hindi / Second Language.
social_science_score	DECIMAL(5,2)	Yes	NULL	Score (0.00 to 100.00) in Social Science.
history_score	DECIMAL(5,2)	Yes	NULL	Score (0.00 to 100.00) in History & Civics.
geography_score	DECIMAL(5,2)	Yes	NULL	Score (0.00 to 100.00) in Geography.
political_science_score	DECIMAL(5,2)	Yes	NULL	Score (0.00 to 100.00) in Political Science.
economics_score	DECIMAL(5,2)	Yes	NULL	Score (0.00 to 100.00) in Economics.
accountancy_score	DECIMAL(5,2)	Yes	NULL	Score (0.00 to 100.00) in Accountancy.
business_studies_score	DECIMAL(5,2)	Yes	NULL	Score (0.00 to 100.00) in Business Studies / Commerce.
psychology_score	DECIMAL(5,2)	Yes	NULL	Score (0.00 to 100.00) in Psychology.
overall_percentage	DECIMAL(5,2)	Yes	NULL	Computed arithmetic mean across entered subjects.
created_at	TIMESTAMP	Yes	CURRENT_TIMESTAMP	Record creation timestamp.

Table 4: question_sections — Standardized test section definitions (5 Columns)

Column Name	Data Type & Constraints	Null	Default	Functional Role & Description
id	INT AUTO_INCREMENT PRIMARY KEY	No	None	Section identifier.
name	VARCHAR(100) NOT NULL	No	None	Standardized section name (e.g. Cognitive Abilities).
description	TEXT	Yes	NULL	Detailed section scope and instructions.
display_order	INT	No	0	UI rendering sequence order.
is_active	BOOLEAN	Yes	TRUE	Active section visibility flag.

Table 5: questions — Adaptive question bank with grade & stream filters (15 Columns)

Column Name	Data Type & Constraints	Null	Default	Functional Role & Description
id	BIGINT UNSIGNED AI PK	No	None	Question master primary key.
question_code	VARCHAR(50) UNIQUE INDEX	No	None	Alphanumeric code (e.g. Q-COG-001).
question_text	TEXT	No	None	Full prompt text rendered to student.
section_id	INT FK INDEX	No	None	FK to question_sections.id (RESTRICT).
question_type	ENUM('rating_scale', 'mcq', 'likert_5', 'likert_7')	No	'mcq'	Interactive UI input component type.
class_min	TINYINT UNSIGNED INDEX	No	7	Minimum student grade level allowed.
class_max	TINYINT UNSIGNED INDEX	No	12	Maximum student grade level allowed.
difficulty	ENUM('easy', 'medium', 'hard')	No	'medium'	Psychometric item difficulty parameter.
skill_category	VARCHAR(100) INDEX	Yes	NULL	Target aptitude or RIASEC interest domain.
stream_specific	VARCHAR(50)	Yes	NULL	Optional stream filter (e.g. Science, Commerce).
is_required	BOOLEAN	Yes	TRUE	Mandatory submission constraint.
display_order	INT	No	0	Section rendering sequence index.
explanation	TEXT	Yes	NULL	Educational explanation for feedback.
is_active	BOOLEAN	Yes	TRUE	Active catalog status flag.
created_at	TIMESTAMP	Yes	CURRENT_TIMESTAMP	Question creation timestamp.

Table 6: question_options — Selectable answer options with psychometric scoring weights (7 Columns)

Column Name	Data Type & Constraints	Null	Default	Functional Role & Description
id	BIGINT UNSIGNED AI PK	No	None	Option primary key.
question_id	BIGINT UNSIGNED FK INDEX	No	None	FK to questions.id (CASCADE DELETE).
option_text	VARCHAR(500)	No	None	Option label displayed to student.
option_value	VARCHAR(100)	Yes	NULL	Normalized code or numeric representation.
score	DECIMAL(5,2)	Yes	0.00	Psychometric weight contributed to ability/interest.
is_correct	BOOLEAN	Yes	NULL	Correctness indicator for cognitive aptitude items.
display_order	INT	No	0	Horizontal/vertical UI arrangement order.

Table 7: assessment_sessions — Assessment attempt state machine & completion tracking (9 Columns)

Column Name	Data Type & Constraints	Null	Default	Functional Role & Description
id	BIGINT UNSIGNED AI PK	No	None	Session primary key.
student_id	BIGINT UNSIGNED FK INDEX	No	None	FK to students.id (CASCADE DELETE).
status	ENUM('in_progress', 'completed', 'expired')	No	'in_progress'	Session lifecycle state token.
started_at	TIMESTAMP	Yes	CURRENT_TIMESTAMP	Test initiation timestamp.
completed_at	TIMESTAMP	Yes	NULL	Final submission timestamp.
current_question	INT	Yes	1	Resume bookmark index for student.
completion_percentage	DECIMAL(5,2)	Yes	0.00	Progress bar percentage (0.00% to 100.00%).
selected_question_ids	TEXT	Yes	NULL	JSON array of deterministic question IDs.
created_at	TIMESTAMP	Yes	CURRENT_TIMESTAMP	Session row creation timestamp.

Table 8: student_answers — Individual student question responses & timing logs (9 Columns)

Column Name	Data Type & Constraints	Null	Default	Functional Role & Description
id	BIGINT UNSIGNED AI PK	No	None	Answer record primary key.
assessment_id	BIGINT UNSIGNED FK INDEX	No	None	FK to assessment_sessions.id (CASCADE DELETE).
question_id	BIGINT UNSIGNED FK INDEX	No	None	FK to questions.id (RESTRICT).
selected_option_id	BIGINT UNSIGNED FK	Yes	NULL	FK to question_options.id (CASCADE DELETE).
selected_option	TEXT	Yes	NULL	Captured response string or option code.
answer_text	TEXT	Yes	NULL	Open text response (if applicable).
numeric_value	DECIMAL(8,3)	Yes	NULL	Direct numeric rating or response value.
time_taken_seconds	INT	Yes	NULL	Response latency for speed/accuracy profiling.
answered_at	TIMESTAMP	Yes	CURRENT_TIMESTAMP	Timestamp of response capture.

Table 9: assessment_scores — Normalized 0-100 Cognitive Aptitudes & RIASEC Vocational Interests (25 Columns)

Column Name	Data Type & Constraints	Null	Default	Functional Role & Description
id	BIGINT UNSIGNED AI PK	No	None	Assessment score primary key.
assessment_id	BIGINT UNSIGNED FK UNIQUE INDEX	No	None	FK to assessment_sessions.id (1:1 CASCADE).
mathematical_ability	DECIMAL(5,2)	Yes	NULL	Cognitive: Mathematical & quantitative reasoning (0-100).
logical_reasoning	DECIMAL(5,2)	Yes	NULL	Cognitive: Deductive & inductive logic (0-100).
scientific_reasoning	DECIMAL(5,2)	Yes	NULL	Cognitive: Empirical scientific methodology (0-100).
problem_solving	DECIMAL(5,2)	Yes	NULL	Cognitive: Complex multi-step problem solving (0-100).
analytical_ability	DECIMAL(5,2)	Yes	NULL	Cognitive: Pattern decomposition & critical analysis (0-100).
communication	DECIMAL(5,2)	Yes	NULL	Cognitive: Verbal & written articulation (0-100).
creativity	DECIMAL(5,2)	Yes	NULL	Cognitive: Divergent thinking & innovation (0-100).
digital_ability	DECIMAL(5,2)	Yes	NULL	Cognitive: Computational thinking & tech fluency (0-100).
learning_ability	DECIMAL(5,2)	Yes	NULL	Cognitive: Conceptual agility & learning rate (0-100).
memory	DECIMAL(5,2)	Yes	NULL	Cognitive: Working & associative memory capacity (0-100).
observation	DECIMAL(5,2)	Yes	NULL	Cognitive: Attention to detail & spatial awareness (0-100).
spatial_ability	DECIMAL(5,2)	Yes	NULL	Cognitive: 2D/3D mental manipulation (0-100).
practical_ability	DECIMAL(5,2)	Yes	NULL	Cognitive: Hands-on execution & kinesthetic skill (0-100).
teamwork	DECIMAL(5,2)	Yes	NULL	Behavioral: Collaborative synergy & interpersonal skill (0-100).
leadership	DECIMAL(5,2)	Yes	NULL	Behavioral: Initiative & decision-making responsibility (0-100).
technology_interest	DECIMAL(5,2)	Yes	NULL	Vocational: Computing, engineering, hardware (0-100).
science_interest	DECIMAL(5,2)	Yes	NULL	Vocational: Natural sciences, biology, chemistry, physics (0-100).
healthcare_interest	DECIMAL(5,2)	Yes	NULL	Vocational: Medicine, clinical care, public health (0-100).
business_interest	DECIMAL(5,2)	Yes	NULL	Vocational: Commerce, finance, entrepreneurship (0-100).
creative_interest	DECIMAL(5,2)	Yes	NULL	Vocational: Arts, design, literature, media (0-100).
research_interest	DECIMAL(5,2)	Yes	NULL	Vocational: Academic exploration & scientific inquiry (0-100).
social_interest	DECIMAL(5,2)	Yes	NULL	Vocational: Community service, psychology, counseling (0-100).
created_at	TIMESTAMP	Yes	CURRENT_TIMESTAMP	Score calculation timestamp.

Table 10: career_domains — Primary industry macro-domains (33 Domains, 6 Columns)

Column Name	Data Type & Constraints	Null	Default	Functional Role & Description
id	INT AUTO_INCREMENT PRIMARY KEY	No	None	Domain master primary key.
domain_name	VARCHAR(150) UNIQUE INDEX	No	None	Standardized sector name (e.g. Technology, Engineering).
description	TEXT	Yes	NULL	High-level industry overview and economic scope.
icon	VARCHAR(100)	Yes	'bi-briefcase'	Bootstrap icon class identifier for UI cards.
display_order	INT	No	0	Visual layout sorting position.
is_active	BOOLEAN	Yes	TRUE	Active domain catalog visibility flag.

Table 11: career_subdomains — Disciplinary sub-sectors (389 Subdomains, 4 Columns)

Column Name	Data Type & Constraints	Null	Default	Functional Role & Description
id	INT AUTO_INCREMENT PRIMARY KEY	No	None	Subdomain primary key.
domain_id	INT FK INDEX	No	None	FK to career_domains.id (RESTRICT).
name	VARCHAR(150) INDEX	No	None	Subdomain title (e.g. Artificial Intelligence, Cardiology).
description	TEXT	Yes	NULL	Specialized field summary.

Table 12: career_clusters — Occupational clusters (466 Clusters, 4 Columns)

Column Name	Data Type & Constraints	Null	Default	Functional Role & Description
id	INT AUTO_INCREMENT PRIMARY KEY	No	None	Cluster primary key.
subdomain_id	INT FK INDEX	No	None	FK to career_subdomains.id (SET NULL).
name	VARCHAR(150) INDEX	No	None	Cluster name (e.g. Machine Learning Engineering).
description	TEXT	Yes	NULL	Cluster profile description.

Table 13: careers — Master career catalog with 2,259 professions (16 Columns)

Column Name	Data Type & Constraints	Null	Default	Functional Role & Description
id	BIGINT UNSIGNED AI PK	No	None	Career master primary key.
career_code	VARCHAR(50) UNIQUE INDEX	No	None	Unique career code (e.g. TECH_AI_001).
career_name	VARCHAR(200) INDEX	No	None	ML Feature: Official job title.
domain_id	INT FK INDEX	No	None	FK to career_domains.id.
subdomain_id	INT FK	Yes	NULL	FK to career_subdomains.id.
cluster_id	INT FK INDEX	Yes	NULL	FK to career_clusters.id.
description	TEXT	Yes	NULL	Comprehensive role responsibilities & day-to-day work.
minimum_education	VARCHAR(150)	Yes	NULL	Baseline degree (e.g., Bachelor of Technology).
typical_education	VARCHAR(150)	Yes	NULL	Standard degree (e.g., Master of Science).
work_environment	VARCHAR(200)	Yes	NULL	Work setting (Office / Remote / Field / Lab).
work_style	VARCHAR(200)	Yes	NULL	Work habits (Analytical, Creative, Collaborative).
entry_level_role	VARCHAR(200)	Yes	NULL	First professional starting position.
advanced_role	VARCHAR(200)	Yes	NULL	Senior milestone position.
related_careers	TEXT	Yes	NULL	Related professions for cross-recommendations.
is_active	TINYINT(1) INDEX	Yes	1	Active catalog visibility flag.
created_at / updated_at	TIMESTAMP	Yes	CURRENT_TIMESTAMP	Authoring and modification timestamps.

Table 14: career_skills — Skill requirements & importance weights (5 Columns)

Column Name	Data Type & Constraints	Null	Default	Functional Role & Description
id	BIGINT UNSIGNED AI PK	No	None	Skill link primary key.
career_id	BIGINT UNSIGNED FK INDEX	No	None	FK to careers.id (CASCADE DELETE).
skill_name	VARCHAR(150)	No	None	Skill name (e.g., Python, Critical Thinking).
importance_level	TINYINT UNSIGNED	No	4	Weight rating (1 = Lowest, 5 = Critical).
importance_label	VARCHAR(30)	Yes	'High'	Text label (Critical, High, Medium).

Table 15: career_subjects — High school subject prerequisites (5 Columns)

Column Name	Data Type & Constraints	Null	Default	Functional Role & Description
id	BIGINT UNSIGNED AI PK	No	None	Subject link primary key.
career_id	BIGINT UNSIGNED FK INDEX	No	None	FK to careers.id (CASCADE DELETE).
subject_name	VARCHAR(150)	No	None	Subject name (e.g., Mathematics, Physics).
importance_level	TINYINT UNSIGNED	No	4	Relevance rating (1 to 5).
importance_label	VARCHAR(30)	Yes	'High'	Text label (Critical, High, Medium).

Table 16: career_education — Higher education degrees & milestones (6 Columns)

Column Name	Data Type & Constraints	Null	Default	Functional Role & Description
id	BIGINT UNSIGNED AI PK	No	None	Education stage primary key.
career_id	BIGINT UNSIGNED FK INDEX	No	None	FK to careers.id (CASCADE DELETE).
education_level	VARCHAR(150)	No	None	Stage level (Undergraduate, Postgraduate, Doctorate).
degree_name	VARCHAR(200)	No	None	Official degree title (e.g. B.Tech Computer Science).
description	TEXT	Yes	NULL	Degree scope & entry requirements.
sequence_order	INT	No	1	Chronological milestone sequence order.

Table 17: career_pathways — Professional career ladder stages (5 Columns)

Column Name	Data Type & Constraints	Null	Default	Functional Role & Description
id	BIGINT UNSIGNED AI PK	No	None	Pathway stage primary key.
career_id	BIGINT UNSIGNED FK INDEX	No	None	FK to careers.id (CASCADE DELETE).
stage_number	INT	No	1	Hierarchical career ladder step (1, 2, 3, 4, 5).
stage_name	VARCHAR(150)	No	None	Role title (Junior Engineer -> Tech Lead -> CTO).
description	TEXT	Yes	NULL	Experience expectations & responsibilities.

Table 18: career_recommendations — Final personalized Top-K ML recommendations (9 Columns)

Column Name	Data Type & Constraints	Null	Default	Functional Role & Description
id	BIGINT UNSIGNED AI PK	No	None	Recommendation primary key.
assessment_id	BIGINT UNSIGNED FK INDEX	No	None	FK to assessment_sessions.id (CASCADE DELETE).
career_id	BIGINT UNSIGNED FK	No	None	FK to careers.id (CASCADE DELETE).
rank_position	INT INDEX	No	1	Top-K Rank (Rank 1 through 10).
score	DECIMAL(8,5)	Yes	NULL	Calibrated ML match probability (0.00% to 100.00%).
recommendation_reason	TEXT	Yes	NULL	Natural language justification generated by AI.
strengths	TEXT	Yes	NULL	Student's highest-matching cognitive/interest traits.
skill_gaps	TEXT	Yes	NULL	Identified development areas for the student.
created_at	TIMESTAMP	Yes	CURRENT_TIMESTAMP	Recommendation generation timestamp.

5. Machine Learning Feature Pipeline Integration

The machine learning engine processes 19 engineered features generated entirely from these normalized database tables. The table below illustrates the exact mathematical transformations and source fields.

ML Input Feature	Source Database Table & Field	Feature Type	Mathematical Transformation / Formulation
age	students.age	Numerical	Direct student age bounded to [10, 25]
class	students.class_level	Numerical	Direct class grade level bounded to [7, 12]
stream	students.stream	Categorical	Ordinal encoding of high school stream specialization
career_name	careers.career_name	Categorical	Ordinal encoding across 2,259 unique job titles
career_domain	career_domains.domain_name	Categorical	Ordinal encoding across 33 industry sectors
career_subdomain	career_subdomains.name	Categorical	Ordinal encoding across 389 disciplinary subdomains
career_cluster	career_clusters.name	Categorical	Ordinal encoding across 466 occupational clusters
ability_match_component (A)	assessment_scores (15 Abilities)	Numerical	Cosine similarity between student cognitive vector & career requirement
interest_match_component (I)	assessment_scores (7 Interests)	Numerical	Domain RIASEC interest similarity matching target career
academic_match_component (Ac)	academic_scores (17 Subjects)	Numerical	Weighted subject performance score matching required career subjects
learning_match_component (L)	assessment_scores.learning_ability	Numerical	Normalized cognitive agility score (0 to 100)
composite_alignment_index	Engineered Domain Metric	Numerical	$0.45 * A + 0.35 * I + 0.10 * Ac + 0.10 * L$
ability_interest_synergy	Engineered Domain Metric	Numerical	$(A * I) / 100.0$
ability_interest_gap	Engineered Domain Metric	Numerical	$ A - I $
min_core_match	Engineered Domain Metric	Numerical	$\min(A, I)$
max_core_match	Engineered Domain Metric	Numerical	$\max(A, I)$
harmonic_core_match	Engineered Domain Metric	Numerical	$2.0 * (A * I) / (A + I + 1e-5)$
geometric_core_synergy	Engineered Domain Metric	Numerical	$\text{sqrt}(\max(0, A * I))$
holistic_synergy	Engineered Domain Metric	Numerical	$(A * I * Ac * L)^{0.25}$

6. Security, Privacy & Referential Integrity Policies

- **Password Security:** Passwords are never stored in plaintext; salted hashes (Script / PBKDF2) protect user identity.
- **Cascade Integrity:** Deleting a student profile cleanly cascades and deletes associated academic scores, sessions, answers, and recommendation logs.
- **Domain Protection:** Core career domains, subdomains, and question sections enforce ON DELETE RESTRICT to prevent orphan taxonomy records.
- **Target Leakage Protection:** The database explicitly excludes compatibility_score and compatibility_label from all feature pipelines.

Database Status: VERIFIED ACTIVE (18 Tables)	Test Suite: 83/83 Tests Passed (OK)	Engine: MySQL 8.x InnoDB (utf8mb4)
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